PVP 23

Code: 23BS1305

II B.Tech - I Semester – Regular / Supplementary Examinations
NOVEMBER 2025

DISCRETE MATHEMATICS AND GRAPH THEORY

(Common for CSE, IT, AIML, DS)

Duration: 3 hours Max. Marks: 70

Note: 1. This question paper contains two Parts A and B.
2. Part-A contains 10 short answer questions. Each Question carries 2 Marks.
3. Part-B contains 5 essay questions with an internal choice from each unit. Each Question carries 10 marks.
4. All parts of Question paper must be answered in one place.

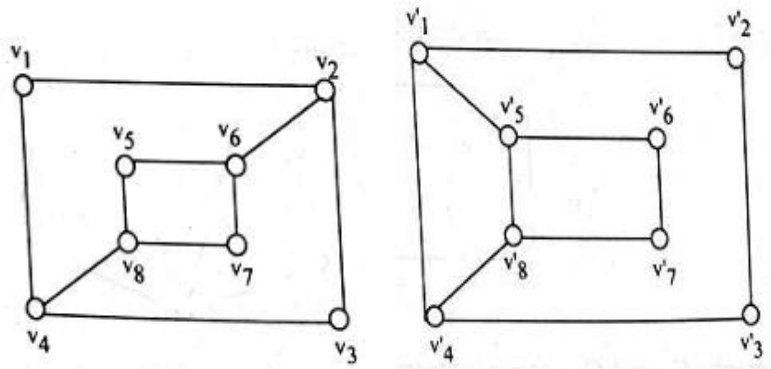
BL – Blooms Level CO – Course Outcome

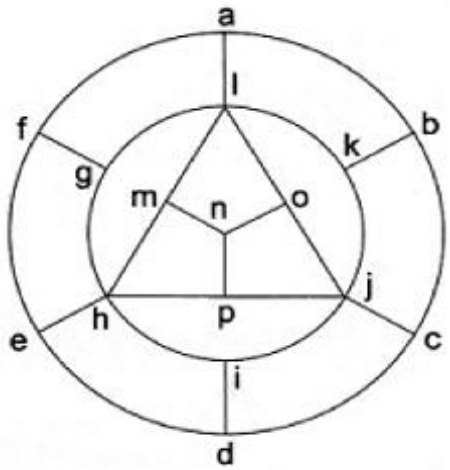
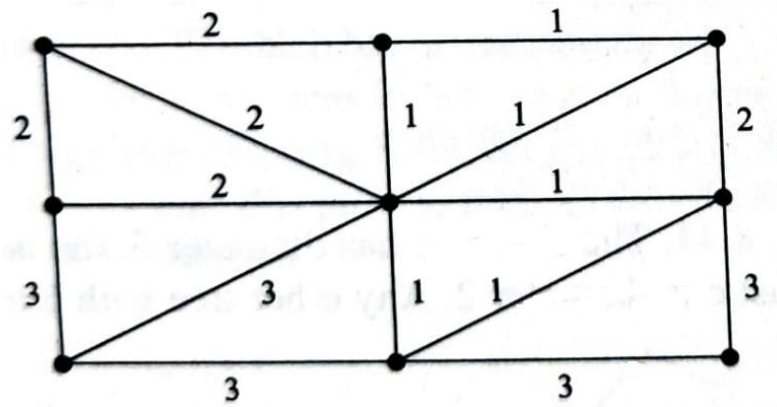
PART – A

		BL	CO
1.a)	Find the duality of the statement $(p \to q) \to r$.	L2	CO1
1.b)	Obtain Conjunctive Normal Form (CNF) of $\neg[p \to (q \vee r)]$.	L2	CO1
1.c)	Define Predicate with an example.	L2	CO1
1.d)	State Rules of Modus Ponens and Modus Tollens.	L2	CO1
1.e)	Write the general form of a third order non-homogeneous recurrence relation.	L2	CO1
1.f)	Solve the recurrence relation $a_n - 4a_{n-1} + 4a_{n-2} = 0$.	L3	CO3
1.g)	Define partially ordered set.	L2	CO1
1.h)	Draw the Hasse diagram for positive divisors of 12.	L2	CO1
1.i)	Define planar graph with an example.	L2	CO1
1.j)	Define chromatic number of a graph.	L2	CO1

PART – B

			BL	CO	Max. Marks
		UNIT-I			
2	a)	Prove that $[P\to(Q\to R)]\to[(P\to Q)\to(P\to R)]$ is a tautology.	L3	CO2	5 M
	b)	Obtain PCNF of $\sim(P\vee Q) \leftrightarrow (P\wedge Q)$.	L3	CO2	5 M
		OR			
3	a)	Prove that $\square(P\wedge Q)\to[\square P\vee(\square P\vee Q)]\Leftrightarrow(\square P\vee Q)$.	L3	CO2	5 M
	b)	Obtain PDNF of $(\square P\to Q)\wedge(Q\leftrightarrow P)$.	L3	CO2	5 M
		UNIT-II			
4	a)	Show that $R\wedge(P\vee Q)$ is a valid conclusion from the premises $P\vee Q,\ Q\to R,\ P\to M$ and $\square M$.	L3	CO2	5 M
	b)	Verify the validity of the following argument: Tigers are dangerous animals. There are tigers. Therefore, there are dangerous animals.	L3	CO2	5 M
		OR			
5	a)	Write the following quantified statements in symbolic form: i. Some roses are yellow ii. All Russians are taller than all Americans iii. Some monkeys have no tail.	L2	CO2	5 M

	b)	Prove that $(\forall x)(P(x)\vee(\forall x)(Q(x) \Leftrightarrow (\forall x)(P(x)\vee Q(x))$ is logically valid.	L3	CO2	5 M
		UNIT-III			
6		Slove $a_{n+2}-5a_{n+1}+6a_n = 3n+4$ with $a_0 = 2,\ a_1 = 5$	L3	CO3	10 M
		OR			
7		Solve $a_r-7a_{r-1}+16a_{r-2}-12a_{r-3} = 0,\ r \geq 3$ with $a_0 = 1,\ a_1 = 4$ and $a_2 = 8$ by the method of characteristic roots.	L3	CO3	10 M
		UNIT-IV			
8		Define an equivalence relation. Let $\mathbb{Z}$ denote the set of integers and the relation R on $\mathbb{Z}$ be defined by aRb if and only if $a-b$ is an integer. Prove that R is an equivalence relation.	L3	CO4	10 M
		OR			
9	a)	Let R be a relation defined on $A=\{1,2,3,4,5\}$ by $R=\{(x,y)/x \geq y\}$. Verify whether R is partial ordered relation or not?	L4	CO4	5 M
	b)	Verify whether the following graphs are isomorphic or not? 	L4	CO4	5 M

UNIT-V					
10	a)	Explain Depth First Search algorithm with an example.	L4	CO4	5 M
	b)	Show that the following graph is not Hamiltonian. 	L4	CO4	5 M
OR					
11	Define minimal spanning tree. Find the minimal spanning tree for the following weighted graph using Krushkal's algorithm: 		L4	CO4	10 M

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II B.Tech - I Semester – Regular Examinations - DECEMBER 2024

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BL – Blooms Level CO – Course Outcome

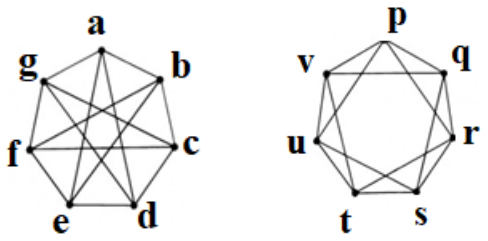
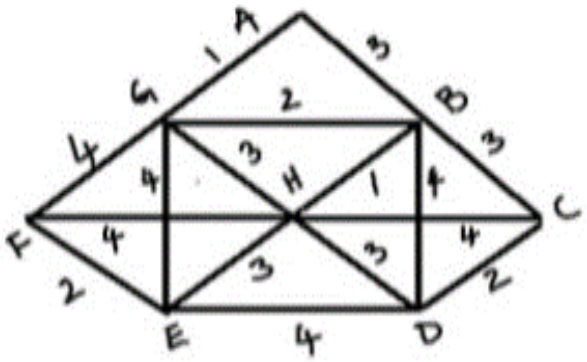
PART – A

		BL	CO
1.a)	Define Proposition.	L1	CO1
1.b)	What is the Difference between CNF and PCNF.	L1	CO1
1.c)	Let Q(x) be the statement "x < 2." What is the truth value of the quantification ∀xQ(x), where the domain consists of all real numbers?	L1	CO2
1.d)	Explain existential quantifier.	L1	CO2
1.e)	Define non-homogeneous recurrence relation of order three.	L2	CO3
1.f)	Solve $a_n + 4a_{n-1} = 2$.	L2	CO3
1.g)	Write Warshall's Algorithm.	L2	CO4
1.h)	Define a Directed Graph.	L1	CO4
1.i)	Define a minimal spanning tree.	L1	CO4
1.j)	Define Hamiltonian Graph.	L1	CO4

PART – B

			BL	CO	Max. Marks
UNIT-I					
2	a)	Show that the proposition $(p \vee \neg q) \wedge (\neg p \vee \neg q) \vee q$ is a tautology.	L2	CO1	5 M
	b)	For any three propositions p, q, r, prove that $[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow q) \wedge (p \rightarrow r)]$	L3	CO2	5 M
OR					
3	a)	Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$	L2	CO1	5 M
	b)	Obtain CDNF of the following $P \rightarrow ((P \rightarrow Q) \wedge \neg (\neg Q \vee \neg P))$	L3	CO2	5 M
UNIT-II					
4	a)	Consider these statements "All lions are fierce", "Some lions do not drink coffee", "Some fierce creatures do not drink coffee" Let P(x), Q(x), and R(x) be the statements "x is a lion", "x is fierce" and "x drinks coffee" respectively. Assuming that the domain consists of all creatures express the statement in the argument using quantifiers and P(x), Q(x) and R(x).	L2	CO1	5 M
	b)	Assume that "For all positive integers n, if n is greater than 4, then n^2 is less than 2^n" is true. Use universal modus ponens to show that $100^2 < 2100$.	L3	CO2	5 M
OR					

5	a)	Show that the premises "A student in this class has not read the book," and "Everyone in this class passed the first exam" imply the conclusion "Someone who passed the first exam has not read the book."	L3	CO2	5 M
	b)	Use contraposition show that if x and y are integers and both xy and $x + y$ are even, then both x and y are even.	L2	CO1	5 M
UNIT-III					
6	a)	Solve the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$ with $a_0 = 2$ and $a_1 = 3$.	L2	CO1	5 M
	b)	Solve the recurrence relation of Fibonacci sequence of numbers $F_{n+2} = F_{n+1} + F_n$ for $n \geq 0$ given that $F_0=0$, $F_1=1$.	L3	CO3	5 M
OR					
7	a)	Solve the following recurrence relation using characteristic roots. $a_n + 4a_{n-1} + 6a_{n-2} = 0$ and $a_0 = 2$, $a_1 = -7$.	L3	CO3	5 M
	b)	Solve $a_n - 9a_{n-1} + 26a_{n-2} - 24a_{n-3} = 0$, for $n \geq 3$.	L3	CO3	5 M
UNIT-IV					
8	a)	Draw the Hasse diagram representing the positive divisors of 36.	L2	CO1	5 M
	b)	Show that the following graphs are isomorphic.	L4	CO4	5 M

		 G_1 G_2			
OR					
9	a)	Show that congruence modulo m is an equivalence relation on integers.	L2	CO1	5 M
	b)	Draw the Hasse diagram representing the partial ordering {(a, b)/a divides b} on {1, 2, 3, 4, 6, 8, 12}.	L4	CO4	5 M
UNIT-V					
10	a)	Write about Euler's circuit and Hamiltonian cycle with suitable examples.	L2	CO1	5 M
	b)	Explain DFS algorithm to find spanning tree of a graph with suitable example.	L4	CO4	5 M
OR					
11		Show step by step Kruskal's algorithm on the following connected weighted graph and also calculate sum of the weights of the minimal spanning tree? 	L4	CO4	10 M